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Chapter 8

Static Routing

This chapter tests you on the basics of IP routing and static routes. The ICND1 certification exam expects you to know the basics of configuring a router and also the basics of managing the routing table through static routing. This chapter also gives example questions on topics such as the gateway of last resort (GWLR) and router on a stick — a new topic for ICND1!

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding routing concepts
- Configuring static routing
- Configuring the gateway of last resort
- Configuring router on a stick

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- A static route is one that is manually added to the router's routing table. You can use the `ip route command` to add a static route to the routing table. You can delete a route from the routing table using the `no ip route` command. When troubleshooting routing issues, use the `show ip route` command to view the routing table.
- Routes have an administrative distance associated with them which indicates the trustworthiness of the knowledge of the route. The lower the administrative distance, the more trustworthy the knowledge is.
- The gateway of last resort (GWLR) feature can be configured on your router so that your router will forward all packets that it has no destination route for to a different router. To configure the GWLR, you use the ip route command to add a route to the 0.0.0.0 network.
- You can connect a switch to a router and have the router route traffic through the VLANs using the router on a stick feature. In order to configure router on a stick, you would create sub-interfaces on the router and enable dot1q as the tagging protocol, along with the VLAN ID, for each sub-interface. You would then configure the switch to use the port with the connection to the router as a trunk port so that it can carry all the VLAN traffic across the port.

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Introduction to Routing and Communication

276–284 Choose the best answer(s).

276. What type of memory stores the routing table on Cisco routers?

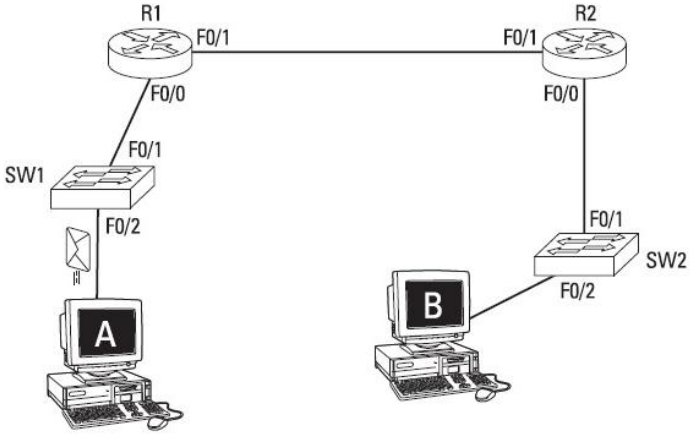
(A) ROM

(B) NVRAM

(C) Flash

(D) RAM

277. Computer A is sending data to computer B on a remote network. Looking at the location of the packet in transit in the figure below, which of the following statements are true? (Select as many as apply.)



(A) The destination IP address is that of Computer B.

(B) The destination IP address is that of router R1.

(C) The destination MAC address is that of the switch.

(D) The source MAC address is that of router R1.

(E) The destination MAC address is that of router R1.

278. When a router receives a packet what does it do? (Select two.)

(A) Determines if a destination route exists in the routing table and what the next hop is

(B) Determines if a destination MAC address appears in the routing table

(C) Looks at the destination IP address in the packet

(D) Filters traffic based on destination MAC address

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279. What routes exist by default on your router?

(A) Static routes

(B) Dynamic routes

(C) Connected routes

(D) Gateway of last resort route

280. Which of the following represents one of the downfalls of static routes?

(A) They are shared automatically with neighboring routers.

(B) They are shared automatically with the entire network.

(C) They are manually configured by the router administrator.

(D) They generate network traffic to broadcast knowledge of the route.

281. Which of the following represents an advantage of static routing?

(A) Less administrative burden over dynamic routing

(B) No network bandwidth is being used by protocols sharing routing tables

(C) Easier to configure on the network than dynamic routing

(D) Uses a higher administrative distance than dynamic routing protocols

282. You are the administrator for router R1 and have configured a static route to the 216.83.11.0 network. Your company has loaded RIPv1 on all the routers on the network, and router R2 shares knowledge of a route to the 216.83.11.0 network with a RIP update. Which route will your router use?

(A) The default route

(B) The static route

(C) The RIPv1 route

(D) The GWLR

283. Which of the following two actions must a router do with an incoming packet in order to send it to its destination? (Choose two.)

(A) Determine if a route exists in the routing table

(B) Look at the source IP address of the packet

(C) Determine if an entry exists in the MAC address table

(D) Look at the destination IP address of the packet

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284. What is the process called that the IP protocol uses to determine if the system it is trying to communicate with is on a different network?

(A) Routing

(B) ANDing

(C) NATing

(D) Switching

Configuring Static Routes

285–294 Choose the best answer(s).

285. What is the command to delete a static route?

(A) `no ip route`

(B) `route delete`

(C) `delete route`

(D) `ip route delete`

286. Your manager has disabled routing on the router by mistake. What command could you use to enable routing again?

(A) `ip routing`

(B) `no ip routing`

(C) `enable routing`

(D) `no disable routing`

287. What is the result of the following commands being entered on the route?

```
RouterA>enable
RouterA#config term
RouterA(config)#ip route 217.56.33.48 255.255.255.240 26.10.20.2
```

(A) Any data destined for the 26.10.20.2/28 network is forwarded to the IP address of 217.46.33.48.

(B) Any data destined for the 217.46.33.48/28 network is forwarded to the IP address of 26.10.20.2.

(C) Any data destined for the 217.46.33.0/28 network is forwarded to the IP address of 26.10.20.2.

(D) Any data destined for the 26.10.20.0/28 network is forwarded to the IP address of 217.46.33.48.

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288. You have configured static routing on your router and would like to remove an entry from the routing table. What command would you use?

(A) `delete 27.0.0.0 255.0.0.0`

(B) `disable 27.0.0.0 255.0.0.0`

(C) `no ip route 27.0.0.0 255.0.0.0 26.0.0.2`

(D) `undo route 27.0.0.0 255.0.0.0`

289. What command adds a static route?

(A) `ip route 35.0.0.0 22.0.0.1`

(B) `route add 35.0.0.0 255.0.0.0 22.0.0.1`

(C) `route add 35.0.0.0 22.0.0.1`

(D) `ip route 35.0.0.0 255.0.0.0 22.0.0.1`

290. You type the following command into the router. Which of the following statements is true as a result of the command?

```
Ip route 200.45.7.0 255.255.255.224 22.202.33.10 10
```

(A) Packets destined for 200.45.7.98 will be forwarded to 22.202.33.10.

(B) The command configures a GWLR of 22.202.33.10.

(C) Packets destined for 200.45.7.45 will be forwarded to 22.202.33.10.

(D) The administrative distance to the destination network is 10.

291. You are the administrator for a small network made up of two routers. What is a quick method you can use to configure routing between all networks?

(A) Build static routes to all unknown routes on each router.

(B) Configure the GWLR on both routers to point to one another.

(C) Configure NAT overload on the first router and Static NAT on the second router.

(D) Configure access lists to allow traffic to route between the networks.

292. Your routers are running the RIP routing protocol and you type the following command. What is the outcome of typing this command?

```
Ip route 0.0.0.0 0.0.0.0 55.12.4.38
```

(A) If there is no matching destination network in the routing table, the router will send the packet to 55.12.4.38.

(B) You have configured RIP to broadcast knowledge of the 55.12.4.38 network.

(C) You have configured a connected route to forward traffic to 55.12.4.38.

(D) You have configured Gateway of Last Resort so that if there is a packet destined for the 55.0.0.0 network, then the router will forward it on to 0.0.0.0.

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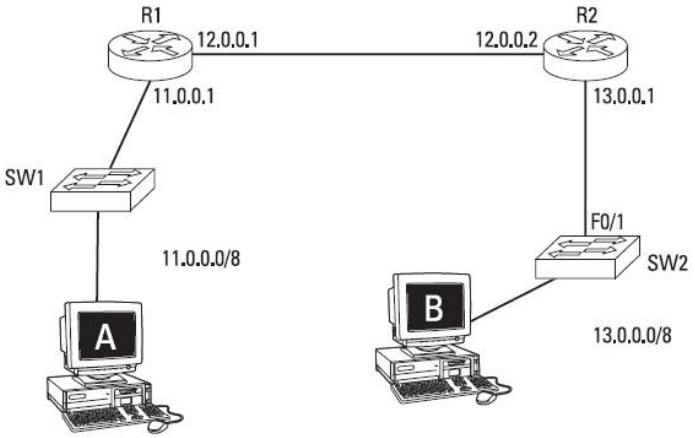
293. You are the administrator for router R1. Looking at the figure below, what command would you use to add a static route the missing network?



(A) `ip route 12.0.0.0 255.0.0.0 12.0.0.1`

(B) `ip route 12.0.0.0 255.0.0.0 11.0.0.1`

(C) `ip route 13.0.0.0 255.0.0.0 12.0.0.1`

(D) `ip route 13.0.0.0 255.0.0.0 12.0.0.2`

294. Tom, one of the network administrators in your office, types the following command into the router. Which of the following statements is true as a result of the command?

```
Ip route 200.45.7.64 255.255.255.224 22.202.33.10 10
```

(A) Packets destined for 200.45.7.98 will be forwarded to 22.202.33.10.

(B) The command configures a GWLR of 22.202.33.10.

(C) Packets destined for 200.45.7.89 will be forwarded to 22.202.33.10.

(D) The hop count to the destination network is 10.

Gateway of Last Resort

295–298 Choose the best answer(s).

295. You are considering configuring default routes on your routers. Which of the following are benefits of default routes? (Select two.)

(A) They allow communication to networks not appearing in the routing table.

(B) The routes never go down.

(C) The routes will take first priority.

(D) The routing table size is kept to a minimum.

(E) They are used to forward traffic from the Internet to your internal network.

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296. Which of the following describes the gateway of last resort?

(A) It is the address of the device on the network that converts FQDNs to IP addresses.

(B) The address that your router will forward a packet to when it does not have a route for that packet.

(C) It is the address of the device that translates private addresses to public addresses.

(D) It is the address of the router that shares routing table information on the network.

297. Your manager asks you what the purpose of a default route is on the router. What would you say?

(A) The default route takes precedence over the static route.

(B) The default route takes precedence over the RIPv2 routes.

(C) The default route is used when there is no other route to the destination.

(D) The default route takes precedence over the RIPv1 route.

298. You would like to configure the GWLR to forward traffic to your ISP_s router, which uses the IP address of 145.66.77.99. What command would you use?

(A) `ip route 145.66.77.99 255.255.255.255 0.0.0.0`

(B) `ip route 145.66.77.99 255.255.0.0 0.0.0.0`

(C) `ip route 145.66.77.99 255.255.0.0 145.66.77.99`

(D) `ip route 0.0.0.0 0.0.0.0 145.66.77.99`

Troubleshooting Connectivity

299–305 Choose the best answer(s).

299. What command is used to view your routing table?

(A) `ip route show`

(B) `show ip route`

(C) `show route table`

(D) `table route show`

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300. What does the following entry in the routing table signify?

S* 0.0.0.0 [1/0] via 56.0.0.1

(A) DNS server

(B) Gateway of last resort

(C) NAT-enabled router

(D) Address of the DHCP enabled router

301. Using the following output, how will data be sent to 26.13.45.222?

```
ROUTER87#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, …
(Additional codes omitted for briefness)
Gateway of last resort is not set
S 29.0.0.0 [1/0] via 26.0.0.2
C 26.0.0.0/8 is directly connected, Serial0/0/0
C 25.0.0.0/8 is directly connected, FastEthernet0/1
```

(A) It will be sent out Fast Ethernet0/1

(B) It will be sent to 26.0.0.2

(C) It will be sent out Serial 0/0/0

(D) It will be sent out Serial 0/0/1

302. You are configuring router on a stick. Which of the following commands would create a sub-interface on the router?

(A) `create sub-interface 20`

(B) `sub-interface fa0/0.20`

(C) `interface fa0/0.20`

(D) `config sub-interface 20`

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303. Using the following output, how will data be sent to 29.66.84.2?

```
ROUTER87#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, …
(Additional codes omitted for briefness)
Gateway of last resort is not set
S 29.0.0.0 [1/0] via 26.0.0.2
C 26.0.0.0/8 is directly connected, Serial0/0/0
C 25.0.0.0/8 is directly connected, FastEthernet0/1
```

(A) Data will be sent out Serial0/0/0.
(B) Data will be sent out FastEthernet0/1.
(C) Data will be sent out Serial0/0/1.
(D) Data will be sent to 26.0.0.2.

304. Using the following output, how will data be sent to 25.33.200.2?

```
ROUTER87#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, …
(Additional codes omitted for briefness)
Gateway of last resort is not set
S 29.0.0.0 [1/0] via 26.0.0.2
C 26.0.0.0/8 is directly connected, Serial0/0/0
C 25.0.0.0/8 is directly connected, FastEthernet0/1
```

(A) Data will be sent out FastEthernet0/1
(B) Data will be sent out Serial0/0/0
(C) Data will be sent out Serial0/0/1
(D) Data will be sent to 26.0.0.2

305. Using the output shown below, which resulted from the show ip route command, what is the hop count to reach the 29.0.0.0 network?

```
S 29.0.0.0 [1/5] via 26.0.0.2
```

(A) 1
(B) 5
(C) 26
(D) 29

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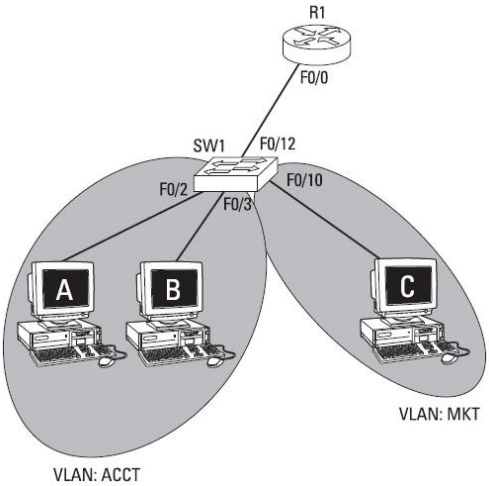
Router on a Stick

306–307 Choose the best answer(s).

306. Which of the following statements are true in order to get traffic to route between the two VLANs in the figure below? (Select two.)



(A) Do nothing.

(B) Configure F0/12 on SW1 for trunk mode.

(C) Configure F0/0 on R1 for trunk mode.

(D) Configure sub-interfaces on SW1.

(E) Configure sub-interfaces on R1.

307. You are configuring router on the stick so the router will route traffic between VLAN 10 and VLAN 20. You configure the sub-interfaces with the following commands. During testing, you notice that the router is not routing between the VLANs. What is missing?

```
interface fa0/0.20
ip address 192.168.20.1 255.255.255.0
interface fa0/0.10
ip address 192.168.10.1 255.255.255.0
```

(A) You need to configure each sub-interface as a trunk port.

(B) You need to configure a different subnet mask on each interface.

(C) You need to enable dot1q and specify the VLAN on each interface.

(D) You need to configure both interfaces for the same network ID.

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Administrative Distances

308–313 Choose the best answer(s).

308. The router will use the route that has the _____ administrative distance.

(A) highest

(B) lowest

309. Your router receives knowledge of a network via RIPv2 that it already has a static route to. Which pathway will be used?

(A) The static route entry

(B) The RIPv2 entry

(C) The default route entry

(D) The packet will be denied

310. What is the administrative distance of RIPv1?

(A) 1

(B) 100

(C) 120

(D) 0

311. What is the administrative distance of a connected route?

(A) 1

(B) 100

(C) 120

(D) 0

312. Which routing feature has an administrative distance of 120?

(A) Static

(B) Connected

(C) RIP

(D) OSPF

313. What is the administrative distance of a static route?

(A) 1

(B) 100

(C) 120

(D) 0